# Netflix Recommendation System Presentation

*(Copy these slides into PowerPoint. Each slide contains a Title, Bullet Points, and Speaker Notes).*

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## Slide 1: Title Slide

**Title**: Personalized Movie Recommendation Engine **Subtitle**: Netflix Prize Dataset Analysis & Modeling **Content**:

- Project Overview
- Collaborative Filtering at Scale
- Model Architecture Comparison

**Speaker Notes**: "Welcome. Today I will present our development of a personalized movie recommendation engine using the famous Netflix Prize Dataset, focusing on how we handled massive data scale and compared neural architectures."

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## Slide 2: Problem Overview & Dataset

**Title**: The Recommendation Challenge **Content**:

- **Massive Scale**: Original dataset contains over 100 million ratings.
- **Extreme Sparsity**: 98%+ sparsity in the raw user-item interaction matrix.
- **Power-Law Distribution**: A few users rate thousands of movies, while most rate very few. A few blockbuster movies receive millions of ratings, while niche titles are ignored.

**Speaker Notes**: "The primary challenge in recommendation systems is data sparsity. Most users have only seen a tiny fraction of the 17,000 available movies. Handling a 100-million row matrix requires intelligent filtering and efficient algorithms."

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## Slide 3: Approach & Methodology

**Title**: Intelligent Filtering & Collaborative Filtering **Content**:

- **Data Pipeline**: Streamed 100M rows and filtered out inactive users (< 500 ratings) and unpopular movies (< 1000 ratings).
- **Optimized Dataset**: Reduced to 45.2M high-quality ratings (0.3GB Parquet format).
- **Methodology**: Adopted Model-Based Collaborative Filtering (Latent Factor Models) rather than Memory-Based (KNN) to overcome memory limitations.

**Speaker Notes**: "By filtering the dataset, we reduced noise and compute costs by 55% while retaining the densest interaction signals. We then used matrix factorization techniques to learn latent representations of users and movies."

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## Slide 4: Model Architecture

**Title**: Deep NCF vs. Generalized Matrix Factorization **Content**:

- **Deep NCF (Neural Collaborative Filtering)**:
    - Embeddings passed through a Multi-Layer Perceptron (MLP).
    - Highly non-linear, 4.0 Million parameters.
- **GMF (Generalized Matrix Factorization)**:
    - Simple dot-product between 128-dimensional user and item vectors.
    - Linear, 7.9 Million parameters.

**Speaker Notes**: "We tested two architectures. Deep NCF uses neural network layers to learn complex patterns, while GMF relies on a mathematically elegant dot-product. We hypothesized that NCF might overfit the sparse data."

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## Slide 5: Experimental Results (RMSE)

**Title**: Rating Prediction Accuracy **Content**:

- **Target**: Achieve an RMSE of ~0.80.
- **Deep NCF Results**:
    - Test RMSE: 0.8123
- **GMF Results**:
    - Test RMSE: **0.7717** (Target Exceeded!)

**Speaker Notes**: "Our results showed that GMF significantly outperformed Deep NCF. The simple dot-product of Matrix Factorization generalized much better to unseen data than the deep MLP layers, successfully beating our 0.80 RMSE target."

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## Slide 6: Experimental Results (MAP@10)

**Title**: Recommendation Ranking Quality **Content**:

- **Metric**: Mean Average Precision at 10 (MAP@10)
- **Relevance Threshold**: Ground truth rating $\geq$ 3.5 stars.
- **Results**:
    - Deep NCF: 0.0743
    - GMF: **0.1673**

**Speaker Notes**: "While RMSE measures prediction error, MAP@10 measures the actual quality of the Top-10 list. GMF was more than twice as effective at surfacing highly relevant, 4 and 5-star movies to the top of the user's feed."

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## Slide 7: Recommendation Examples

**Title**: Real-World Top-10 Output **Content**:

- **Test User**: Profile with high affinity for sci-fi/fantasy.
- **GMF Top Recommendations**:
    1. *Lord of the Rings: Return of the King* (Pred: 5.0)

2. *Star Wars: Episode V* (Pred: 5.0)
- **Success**: Accurate genre clustering and personalization.
- **Failure Case**: "Popularity Bias" — struggles to recommend highly relevant but niche indie films.

**Speaker Notes**: "When we look at the actual recommendations, the model succeeds at capturing user taste clusters. However, due to our initial dataset filtering, it suffers from popularity bias, mostly recommending safe blockbusters."

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## Slide 8: Key Insights & Conclusion

**Title**: Takeaways & Future Work **Content**:

- **Simplicity Wins**: Linear Matrix Factorization (GMF) beats Deep Neural Networks (NCF) on sparse collaborative filtering tasks.
- **Production Readiness**: GMF's dot-product allows for real-time serving using fast vector databases (e.g., FAISS).
- **Future Work**: Implement Hybrid Models using content-metadata (genres, tags) to solve the Cold-Start problem for new users.

**Speaker Notes**: "In conclusion, simpler linear models remain state-of-the-art for sparse matrix completion. For future work, we would integrate content metadata to combat popularity bias and solve the cold-start problem. Thank you."